

Linear Regression: Gradient Descent - Second Iteration Solution

This document provides a comprehensive, step-by-step mathematical walkthrough for the **Second Iteration** of parameter optimization using Gradient Descent for a simple Linear Regression model.

1. Problem Formulation and Initial Parameters

The linear model hypothesis is defined as:

$$y_{\text{pred}} = w * x + b$$

The objective is to minimize the Mean Squared Error (MSE) loss function:

$$L(w, b) = (1 / n) * \text{SUM}((y_{\text{pred}_i} - y_i)^2)$$

Given Parameters for Iteration 2:

- **Weight (w):** 0.182 (updated from Iteration 1)
- **Bias (b):** 0.0645 (updated from Iteration 1)
- **Learning Rate (α):** 0.01
- **Number of data points (n):** 4

Dataset:

Point (i)	Input (x)	Actual Target (y)
1	1.0	2.0
2	2.0	2.8
3	3.0	3.6
4	4.0	4.5

2. Step 1: Compute New Predictions

Using the formula $y_{\text{pred}} = (0.182 * x) + 0.0645$:

- For x = 1: $y_{\text{pred}_1} = (0.182 * 1) + 0.0645 = \mathbf{0.2465}$
- For x = 2: $y_{\text{pred}_2} = (0.182 * 2) + 0.0645 = 0.3640 + 0.0645 = \mathbf{0.4285}$
- For x = 3: $y_{\text{pred}_3} = (0.182 * 3) + 0.0645 = 0.5460 + 0.0645 = \mathbf{0.6105}$
- For x = 4: $y_{\text{pred}_4} = (0.182 * 4) + 0.0645 = 0.7280 + 0.0645 = \mathbf{0.7925}$

3. Step 2: Compute Residual Errors and Gradient Components

Next, we calculate the error term ($y_{\text{pred}} - y$) and the product $(y_{\text{pred}} - y) * x$ for each observation:

i	x_i	y_i	y_pred_i	Error (y_pred_i - y_i)	(y_pred_i - y_i) * x_i
1	1.0	2.0	0.2465	-1.7535	-1.7535
2	2.0	2.8	0.4285	-2.3715	-4.7430
3	3.0	3.6	0.6105	-2.9895	-8.9685
4	4.0	4.5	0.7925	-3.7075	-14.8300
Sum				-10.8220	-30.2950

4. Step 3: Compute Partial Derivatives (Gradients)

A. Gradient with respect to Weight (w):

```
dL/dw = (2 / n) * SUM((y_pred_i - y_i) * x_i)
dL/dw = (2 / 4) * (-30.2950)
dL/dw = 0.5 * (-30.2950) = -15.1475
```

B. Gradient with respect to Bias (b):

```
dL/db = (2 / n) * SUM(y_pred_i - y_i)
dL/db = (2 / 4) * (-10.8220)
dL/db = 0.5 * (-10.8220) = -5.4110
```

5. Step 4: Update the Model Parameters

Using the Gradient Descent update formula with learning rate $\alpha = 0.01$:

Update Weight (w):

```
w_new = w_old - (alpha * dL/dw)
w_new = 0.182 - [0.01 * (-15.1475)]
w_new = 0.182 + 0.151475
w_new = 0.333475 (approx. 0.3335)
```

Update Bias (b):

```
b_new = b_old - (alpha * dL/db)
b_new = 0.0645 - [0.01 * (-5.4110)]
b_new = 0.0645 + 0.05411
b_new = 0.11861 (approx. 0.1186)
```

6. Iteration Comparison Summary

Iteration Stage	Weight (w)	Bias (b)	dL/dw	dL/db
Initial State	0.0000	0.0000	-18.2000	-6.4500
After Iteration 1	0.1820	0.0645	-15.1475	-5.4110
After Iteration 2 (Final)	0.3335	0.1186	-	-